

IB MATH AA SL

Lesson 39

Optimisation Problems

Calculus · 60-minute lesson

Syllabus SL 5.7 · Property of Lynda Badmus Education

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Do Now — recall

ON YOUR WHITEBOARD

1. Differentiate $A = 20x - x^2$.
2. Solve $20 - 2x = 0$.
3. What does $f' = 0$ find?

Answers (click to reveal)

1. $20 - 2x$
2. $x = 10$
3. stationary (max/min) points

What we're learning today

LEARNING OBJECTIVES

- 1 Form**
an equation for the quantity to optimise.
- 2 Differentiate**
and set the derivative to zero.
- 3 Solve**
for the optimal value.
- 4 Confirm**
it is a maximum or minimum.

SUCCESS CRITERIA — I CAN...

- ✓
write the quantity as a function of one variable
- ✓
differentiate and set $f' = 0$
- ✓
solve for the optimal x
- ✓
confirm max/min and answer in context

The optimisation method

STEPS

$$\frac{dA}{dx} = 0$$

1. write the quantity (area, volume, cost...) as a function
2. reduce to ONE variable using any constraint
3. differentiate, set = 0, solve, then check

REAL CONTEXTS

largest area for a fixed fence

cheapest box for a fixed volume

Always

answer the actual question.

Maximise an area

QUESTION

A rectangle's area is $A = 20x - x^2$.

Find the value of x that maximises A .

SOLUTION

Differentiate and set to zero:

$$\frac{dA}{dx} = 20 - 2x = 0$$

Solve:

$$x = 10$$

$f'' = -2 < 0 \rightarrow$ confirms a maximum.

Your turn — optimise

ANSWER IN YOUR BOOK

1. $A = 12x - x^2$. Find x for max A .
2. Then find the maximum area.
3. $C = x^2 - 10x + 30$. Find x for min C .

Answers (click to reveal)

1. $12 - 2x = 0 \rightarrow x = 6$
2. $A = 72 - 36 = 36$
3. $2x - 10 = 0 \rightarrow x = 5$

Confirming & answering

CHECK THEN STATE

$$f''(x) < 0 \Rightarrow \text{max}$$

Confirm

use f'' (or a sign check) to prove max/min

Answer

give the quantity asked for, with units

COMMON SLIP

Stopping at x and forgetting the actual value.

Read

the question: do they want x , or the area/volume?

Find the maximum value

QUESTION

Using $A = 20x - x^2$ with $x = 10$,
find the maximum area.

SOLUTION

Substitute $x = 10$ into A :

$$A = 20(10) - 10^2$$

Evaluate:

$$A = 200 - 100 = 100$$

Exam-style question

A farmer has 40 m of fence for three sides of a rectangle. The area is $A = x(40 - 2x)$.

- (a) Expand A . [1]
- (b) Find x that maximises the area. [3]
- (c) Find the maximum area. [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $A = 40x - 2x^2$ (A1)
- (b) $dA/dx = 40 - 4x = 0$ (M1)(A1) $\rightarrow x = 10$ (A1)
- (c) $A = 40(10) - 2(100) = 200 \text{ m}^2$ (M1)(A1)

Common mistakes



One variable

Use the constraint to remove the second variable.



Confirm max/min

Always justify with f'' or a sign test.



Answer in context

Give area/volume/cost, not just x .



Units

Include them in the final answer.

Mixed practice

QUESTIONS

1. $A = 16x - x^2$. Max at $x = ?$
2. Max area in Q1?
3. $C = x^2 - 6x + 11$. Min at $x = ?$
4. Confirm a max: f'' should be ?

Answers (click to reveal)

1. $16 - 2x = 0 \rightarrow x = 8$
2. $128 - 64 = 64$
3. $x = 3$
4. negative

Plenary & exit ticket

KEY FORMULAS

$$\frac{dA}{dx} = 0$$

$$f''(x) < 0 \Rightarrow \text{max}$$

EXIT TICKET — before you leave

1. $A = 10x - x^2$. Max at $x = ?$

2. Then max A ?

3. Confirm a min: f'' is?

Answers (click to reveal)

1. $x = 5$
2. 25
3. positive

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Optimisation problems.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 40

Kinematics with calculus — motion in a line.

$$v = \frac{ds}{dt}, \quad a = \frac{dv}{dt}$$